

HUALIEN, Taiwan

WMO: 466990

Lat: 23.9751N

Lon: 121.6132E

Elev: 16

StdP: 101.13

Time Zone: 8.00 (E08)

Period: 90-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	12.8	13.8	7.1	6.3	15.0	8.3	6.8	15.5	9.3	18.8	8.4	18.6	2.5	240	0.283

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.8	32.6	27.4	32.0	27.0	31.4	26.8	28.0	31.7	27.6	31.2	27.1	30.8	3.4	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.9	22.6	31.1	26.4	22.0	30.6	26.1	21.5	30.2	89.7	31.8	87.8	31.4	86.0	30.9	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.3	6.2	DB	10.5	34.6	0.9	1.6	9.8	35.7	9.3	36.6	8.8	37.5	8.1	38.7	(4)
(5)			WB	8.7	29.0	1.4	0.7	7.7	29.5	6.9	29.9	6.1	30.3	5.1	30.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.6	18.4	18.3	20.4	22.9	25.1	27.1	28.5	28.3	26.6	24.5	22.6	20.2
	DBStd	4.02	2.26	2.47	2.67	2.62	1.96	1.58	1.09	1.08	1.41	1.60	1.81	2.09
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	82	29	29	12	2	0	0	0	0	0	0	1	8
	CDD10.0	4966	259	232	323	387	469	513	573	567	497	450	378	317
	CDD18.3	2005	29	27	77	139	211	263	314	309	247	192	129	67
	CDH23.3	16774	23	39	230	751	1579	2765	3839	3646	2278	1108	429	86
CDH26.7	4946	0	0	8	97	326	880	1518	1365	566	153	31	2	
Wind	WSAvg	2.4	2.6	2.5	2.5	2.4	2.1	2.2	2.3	2.3	2.4	2.5	2.5	2.7
Precipitation	PrecAvg	2129	67	101	88	111	204	229	179	236	353	357	133	54
	PrecMax	3558	200	338	290	369	600	698	791	679	966	1700	428	143
	PrecMin	989	12	24	24	26	30	26	14	3	32	12	4	15
	PrecStd	556	36	77	50	78	122	157	163	188	238	377	112	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.2	25.8	27.6	30.1	31.0	32.4	33.9	33.6	32.1	30.2	28.5	26.8
		MCWB	20.7	21.9	23.5	25.0	26.4	26.9	28.3	27.7	26.7	25.9	24.1	22.9
	2%	DB	24.0	24.6	26.6	28.8	30.2	31.7	32.9	32.5	31.1	29.2	27.6	25.2
		MCWB	20.1	20.9	22.7	24.5	25.9	26.9	27.6	27.3	26.3	24.9	23.3	21.2
	5%	DB	22.9	23.5	25.7	27.8	29.4	31.1	32.1	31.8	30.3	28.4	26.7	24.2
		MCWB	19.2	20.0	22.2	24.1	25.3	26.7	27.0	27.0	25.8	24.2	22.5	20.5
	10%	DB	21.9	22.2	24.7	26.9	28.5	30.4	31.4	31.2	29.5	27.4	25.7	23.3
		MCWB	18.6	19.1	21.6	23.5	24.8	26.3	26.8	26.8	25.3	23.5	21.7	19.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.5	22.7	24.1	26.0	27.1	27.7	28.8	28.5	27.5	26.7	25.1	23.2
		MCDB	24.4	25.1	26.9	28.8	30.5	31.3	33.1	32.4	31.2	29.6	27.5	26.0
	2%	WB	20.5	21.4	23.3	25.0	26.2	27.4	28.1	27.9	26.8	25.7	24.1	22.1
		MCDB	23.2	24.0	26.0	28.1	29.4	31.0	32.0	31.6	30.2	28.3	26.9	24.5
	5%	WB	19.7	20.4	22.5	24.2	25.7	26.9	27.6	27.5	26.3	24.9	23.2	21.2
		MCDB	22.3	22.8	25.2	27.4	28.7	30.6	31.4	31.1	29.5	27.4	25.9	23.6
	10%	WB	19.0	19.6	21.8	23.7	25.1	26.5	27.2	27.0	25.8	24.2	22.4	20.2
		MCDB	21.5	21.8	24.4	26.7	28.0	29.9	30.8	30.6	28.7	26.4	25.0	22.8
Mean Daily Temperature Range	5% DB	MDBR	4.8	4.6	4.9	5.0	5.2	5.3	5.8	5.7	5.3	4.8	5.0	4.6
		MCDBR	6.6	6.7	6.6	6.6	6.4	6.5	6.8	6.6	6.4	6.3	6.5	6.2
		MCWBR	4.2	4.7	4.2	4.5	4.3	4.4	4.4	4.1	3.9	3.7	4.4	4.0
	5% WB	MCDBR	6.1	6.2	6.1	6.0	5.9	6.1	6.3	6.1	5.8	5.7	6.0	5.3
		MCWBR	4.2	4.6	4.2	4.3	4.1	4.2	4.3	3.9	3.7	3.5	4.6	3.7
Clear-Sky Solar Irradiance	taub	0.544	0.578	0.700	0.706	0.614	0.551	0.528	0.563	0.562	0.546	0.534	0.549	
	taud	1.894	1.853	1.618	1.658	1.883	2.054	2.124	2.009	1.995	1.990	1.996	1.908	
	Ebn at Noon	703	714	650	660	721	762	781	754	745	732	709	678	
	Edh at Noon	178	197	261	255	203	170	158	177	176	169	159	170	
All-Sky Solar Radiation	RadAvg	2.38	2.77	3.15	3.52	4.13	4.92	5.73	5.15	4.51	3.75	2.76	2.35	
	RadStd	0.34	0.45	0.46	0.55	0.68	0.58	0.36	0.51	0.48	0.40	0.29	0.36	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)	N/A	N/A	N/A	+0.27	N/A	N/A	N/A	N/A	N/A	+81

Nomenclature: See separate page